DEPARTMENT OF THE AIR FORCE
HEADQUARTERS UNITED STATES AIR FORCE
WASHINGTON DC

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MEMORANDUM FOR DISTRIBUTION C
MAJCOMs/FLDCOMs/FOAs/DRUs

FROM: AF/A4
1030 Air Force Pentagon, Suite 4E154
Washington, DC 20330-1030

SUBJECT: Department of the Air Force Guidance Memorandum to Department of the Air Force Manual 32-1062, *Electrical Systems, Power Plants and Generators*

By Order of the Secretary of the Air Force, Air Force Manual (AFMAN) 32-1062, *Electrical Systems, Power Plants and Generators*, is re-designated Department of the Air Force Manual (DAFMAN), and this DAF Guidance Memorandum (DAFGM) immediately implements changes to DAFMAN 32-1062. Compliance with this memorandum and its attachments is mandatory. To the extent its direction is inconsistent with other DAF publications, the information herein prevails in accordance with Department of the Air Force Instruction (DAFI) 90-160, *Publications and Forms Management.* Applies to all DAF installations, including Guard and Reserve.

This memorandum clarifies the Base Civil Engineer's (BCE) responsibility to establish and uphold oversight procedures for maintenance of generators assigned to the BCE and to maintain accurate and complete inventory data of those generators. It grants the National Guard Bureau (NGB) authority to oversee generator installations for NGB-related purposes, while also streamlining generator requests through CE DASH; standardizing generator status reporting via the Air Program Information Management System (APIMS); establishing a requirement for BCE approval for all portable generator connection points, extending this approval to generators for facilities listed in **Chapter 3**; and increasing the size limit for portable generators at connection points without authorization for an Equipment Authorized Inventory Data (EAID) generator.

The use of an asterisk (*) identifies a substantive change over the previous guidance. This memorandum becomes void after one year has elapsed from the date of this memorandum, or upon publication of an interim change to or rewrite of DAFMAN 32-1062, whichever is earlier.

ANTHONY R. BAITY, SES, DAF
Acting DCS/Logistics, Engineering & Force Protection

Attachment:
DAFMAN32-1062_DAFGM2025-01, *Electrical Systems, Power Plants and Generators*

ATTACHMENT 1
Electrical Systems, Power Plants and Generators

Current guidance in DAFMAN 32-1062, *Electrical Systems, Power Plants and Generators*, remains in effect with the following changes:

Attachment 1

*Opening Paragraph. **Changed**. This Department of the Air Force Manual (DAFMAN) implements Air Force Policy Directive (AFPD) 32-10, *Installations and Facilities*. It provides acquisition and operation and maintenance (O&M) requirements for power systems, individual Real Property Installed Equipment (RPIE), and Equipment Authorized Inventory Data (EAID) generators. It applies to the entire DAF, including DAF civilian employees and uniformed members of the United States Air Force (USAF), the United States Space Force (USSF or Space Force), the Air Force Reserve (AFR), the Air National Guard (ANG), and military and civilian personnel (including tenants and contractors) performing work or inspections in accordance with this manual, except where noted otherwise. Ensure all records generated because of processes prescribed in this publication adhere to Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with the Air Force Records Disposition Schedule, which can be referenced in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the office of primary responsibility (OPR) using the DAF Form 847, *Recommendation for Change of Publicatio*n; route DAF Forms 847 from the field through the appropriate functional chain of command. This publication may be supplemented at any level, but all supplements must be routed to the OPR of this publication for coordination prior to certification and approval. The authorities to waive wing, unit, delta, or garrison level requirements in this publication are identified with a tier ("T-0, T-1, T-2, T-3") number following the compliance statement. Submit requests for waivers through the chain of command to the appropriate tier waiver approval authority, or alternately, to the publication OPR for non-tiered compliance items. See DAFMAN 90-161, *Publishing Processes and Procedures*, for a description of the authorities associated with the tier numbers. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Department of the Air Force.

*1.2.2. **Changed**. Develop, maintain, clarify, and publish guidance for electrical systems, power plants, and generator programs in coordination with SAF/IEE.

***1.3. Changed. Major Commands (MAJCOM) and Field Commands (FLDCOM).**

*1.4.1. **Changed.** Review and coordinate on generator authorization and design requests or recommended alternatives for AFRC installations. AFRC can submit requests for technical support to the Air Force Civil Engineer Center, Operations Directorate (AFCEC/CO). Process generator authorization and design requests for AFRC installations.

*1.4.2. **Changed**. Review annual RPIE generator inventory and authorization reports, the list of generators that are oversized, and the list of those available for relocation.

*1.5.1. **Changed**. Review and coordinate on generator authorization and design requests or recommended alternatives for ANG installations. ANG can submit requests for technical support to AFCEC/CO to process generator authorization and design requests for ANG. Process generator authorization and design requests for ANG installations.

*1.5.2. **Changed**. Review annual RPIE generator inventory and authorization reports, the list of generators that are oversized, and the list of those available for relocation.

*1.7.1.1. **Changed**. For generator installations that are not listed in **paragraphs 3.1.1** through **3.1.34**, process generator authorizations, designs, and classifications, to include Critical Operations Power Systems (COPS).

*1.7.1.2. **Deleted.**

*1.7.1.3. **Changed**. Approve use of generators for prime power, co-generation, or microgrids. Document all microgrids using the generator authorization module of the Generator Authorization Tool within the CE DASH portal: https://usaf.dps.mil/sites/CE-DASH-Tools/GenAuthorization/Module/Home.aspx?page=Home. Reference DAFI 90-1701, *Installation Energy and Water Management*, for the most up-to-date policy; however, as of 17 December 2020, the DAF's preference is for primary power generation systems to be owned and operated by third parties. Should an installation commander recommend the development of a DAF-owned primary power generation system, the installation commander must obtain a waiver prior to project development from the office of the Deputy Assistant Secretary of the Air Force for Infrastructure, Energy, and Environment (SAF/IEE).

*1.7.2. **Changed**. Maintain the master list of RPIE generators available for relocation as compiled from AFCEC/COM and documented on the CEMIRT home page within the CE DASH portal: https://usaf.dps.mil/teams/CEDASH/SitePages/Service%20Page.aspx?Service=Civil%20Engineer%20Maintenance%20Inspection%20and%20Repair%20Team%20(CEMIRT).

*1.7.6. **Changed**. For generator installations on facilities not listed in **paragraph 3.1.1** through **3.1.34**, coordinate authorization and designs through AFCEC/CO when projects are programmed without an existing generator authorization.

***1.8. Changed. Installation/Center/Space Base Delta Commander.** As the Responsible Official, be the approving authority for deviations from manufacturer service intervals and ensure compliance with all applicable air quality permits. The list of applicable laws includes, but is not limited to: Title 40 Code of Federal Regulations (CFR), Part 63 Subpart ZZZZ, *National Emissions Standards for Hazardous Air Pollutants for Stationary Reciprocating Internal Combustion Engines*; 40 CFR Part 60 Subpart JJJJ, New Source Performance Standards (NSPS), *Standards of Performance for Stationary Spark Ignition Internal Combustion Engines* (the "spark ignition NSPS rule"); 40 CFR Part 60 Subpart IIII, *Standards of Performance for Stationary Spark Ignition Internal Combustion Engines* (the "Compression Ignition NSPS rule"); 40 CFR Part 1039, *Control Of Emissions from New and In-Use Nonroad Compression-Ignition*

Engines; 40 CFR Part 90, *Control of Emissions From Nonroad Spark-Ignition Engines at or Below 19 Kilowatts*; 40 CFR Part 1048, *Control of Emissions from New Large Nonroad Spark-Ignition Engines*.

*1.9.1. **Changed**. Provide, operate, and maintain all real property, RPIE electrical power, and related systems, such as RPIE and EAID items assigned to the BCE. This does not apply to units supporting missile systems or special-use tenant generators, except in cases where there is a signed MOA. The BCE is responsible for ensuring that preventive maintenance procedures comply with DAFI 32-1001, *Civil Engineer Operations*, and that testing frequencies meet the requirements outlined in UFC 3-540-07, *Operation and Maintenance (O&M)*: *Generators*. Additionally, the BCE will operate and maintain other real property and RPIE electrical systems (as applicable to each installation), ensuring all required maintenance tasks are performed according to established schedules and procedures. These other real property and RPIE electrical systems include:

*1.9.4. **Changed**. Approve generators in facilities listed in **paragraph 3.1.1** through **paragraph 3.3.34**, which the BCE is authorized to do. Coordinate all generator sizing designs for RPIE and BCE-owned EAID generators (including generators procured with military construction, O&M, sustainment, restoration, and modernization, DLA, or other funding) through AFCEC/COSM, in accordance with design criteria in UFC 3-540-01, *Engine-Driven Generator Systems for Prime and Standby Power Applications*. Authorizations will be maintained in a generator record log. Memorandum templates and fillable form examples are located on the CE DASH portal: https://usaf.dps.mil/sites/CE-DASH-Tools/GenAuthorization/Module/Home.aspx?page=Home. **Note:** AFRC installations submit to AFRC/A4C for final approval. ANG installations submit to NGB Civil Engineer Technical Branch (CETB) (NGB/A4IC CETB) for processing.

*1.9.5. **Changed**. Establish and maintain oversight procedures to ensure accurate and complete inventory data of generators assigned to the BCE.

*1.9.5.1. **Added**. Oversee an annual inventory, as well as a records update and review, of all CE-operated and maintained RPIE and EAID generators (including replacement-eligible generators and those designated or operated under a National Security Exemption that are exempt from all Environmental Protection Agency (EPA) air permitting requirements) by 1 August each year, in accordance with UFC 3-540-07, Chapter 8. RPIE generators should be reconciled with real property records and BUILDER Sustainment Management System (SMS) records. EAID generators should be reconciled with Custodian Authorization/Custody Receipt Listing (CA-CRL) records.

*1.9.5.2. **Added**. The inventory must be updated in the Facility Power module of the APIMS.

*1.9.5.3. **Added**. The BCE will coordinate with all generator record holders when carrying out this annual inventory, records review, and update process. **Note:** AFRC installations submit to AFRC/A4C for final approval. ANG installations submit to NGB/A4IC CETB for processing.

*1.9.6. **Changed**. Oversee and ensure validation and revalidations for all generators, including updates in APIMS and real property records, are conducted annually and reflect any recent

changes to mission or electrical system configurations. Ensure the section responsible for generator maintenance completes the annual generator inventory and revalidation using the standardized inventory template in the Generator Authorization Tool on the CE DASH portal: https://usaf.dps.mil/sites/CE-DASH-Tools/GenAuthorization/Module/Home.aspx?page=Home. **Note**: AFRC installations submit to AFRC/A4C for final approval. ANG installations submit to NGB/A4IC CETB for processing.

*1.9.7. **Changed**. Ensure compliance with all applicable U.S. federal, state, and local requirements. For installations outside the United States, ensure compliance with the country-specific DoD Final Governing Standards (FGSs). Where no FGSs are established, ensure compliance with DoDM 4715.05, Volumes 1-5, *Overseas Environmental Baseline Guidance Document (OEBGD)*. Ensure generator runtime, maintenance and testing information, and operational data are recorded in APIMS as the system of record (generator runtime; emergency, maintenance/testing, or non-emergency situation).

*1.9.8. **Changed**. Develop and document a prioritized refueling plan for all EAID and RPIE generators, and those under MOAs, to support at least seven days of operation. Plans must be repeatable, and fuel must comply with applicable standards. Liquid petroleum gas (LPG) and natural gas are permitted only in accordance with UFC 3-540-07, Chapter 8.

*1.9.9. **Changed**. The BCE will ensure personnel conducting routine maintenance and physical inventory have appropriately sized and rated fire extinguishers readily available near generators and are trained in their proper use to help protect personnel and equipment.

*1.9.10. **Changed**. Account for generators in accordance with DAFI 32-9005, *Real Property Accountability*; as identified in **paragraph 1.9.5** of this publication; and UFC 3-540-07, Chapter 8, Paragraph 8-8. The more restrictive guidance in **paragraph 1.9.5** of this publication will apply until UFC 3-540-07 is updated from the current version (dated 5 November 2019).

*1.9.12. **Added**. Approve all generator connection points with Manual Transfer Switches (MTS) on all facilities, including Electric Vehicle Charging Facilities (EVCF) not assigned to a critical mission in **Chapter 3** and facilities with a RPIE generator.

*1.9.13. **Added**. Establish a documented maintenance oversight process as part of the annual review of all generator maintenance records. Conduct this review during the annual inventory or revalidation using the standard inventory template in the Generator Authorization Tool on the CE DASH portal: https://usaf.dps.mil/sites/CE-DASH-Tools/GenAuthorization/Module/Home.aspx?page=Home.

*1.9.14. **Added**. Ensure personnel responsible for generator maintenance complete required periodic training, adhering to the applicable Career Field Education and Training Plan (CFETP).

*1.9.15. **Added**. Establish and maintain oversight procedures to verify generator design and sizing approvals are conducted in accordance with UFC 3-540-01 and installed generators are properly sized to meet the intended load.

*****1.9.16. **Added.** Ensure the Self-Assessment Checklist (SAC) corresponding to this DAFMAN is completed and submitted in the Management Internal Control Toolset (MICT) system to verify compliance with all relevant requirements.

*1.11.1. **Changed**. Verify proper function of all building equipment and systems during generator functional testing using facility load, and sign the AF Form 487, *Generator Operating Log (Inspection Checklist)* (or electronic version), following the semi-annual generator test.

***1.12. Changed. Tenant Organizations.** All tenant organizations will submit a work request for any generator system or generator connection point to be connected to real property, and establish an MOA with the BCE for operation and maintenance of the generator on a reimbursable basis. In the absence of an MOA, tenant organizations are responsible for meeting all requirements of this instruction, either through internal resources or unit-funded contract maintenance services.

***2.1. Changed. Prime Power.** AFCEC/CO, in coordination with SAF/IEE, AF/A4C, and AFCEC/CN, manages the use of generators for prime power, microgrids, or co-generation. When submitting a request for co-generation or microgrids within the area electrical power system, the BCE must demonstrate that the system complies with Institute of Electrical and Electronics Engineers (IEEE) 1547, *Standard for Interconnecting Distributed Resources with Electric Power Systems*, and negotiate an interconnection agreement with the area electrical power system (utility) supplier. **(T-2)** Coordination with the BCE's Environmental Section is required prior to authorization for purposes of regulatory permitting and notifications. **(T-3)**

***2.2. Changed. Standby Generators.**

*****2.2.1. **Added.** The BCE will verify authorized mission-essential functions receive standby generator support, with generators installed and configured to supply power to mission-essential loads within a single facility during commercial power loss; for unique circumstances requiring a single standby system to support multiple facilities, the BCE or equivalent commander must request authorization from AFCEC/COSM prior to implementation. **(T-2)**

*****2.2.2. **Added**. Installation Energy Plans or other energy resilience plans may identify opportunities to interconnect multiple standby generators and facilities. In such cases, SAF/IEE may approve projects based on the concept of interconnections. **Note:** AFRC installations submit to AFRC/A4C for final approval. ANG installations submit to NGB/A4IC CETB for processing.

*2.3.1.2. **Changed**. Critical Operations Power Systems (COPS). The BCE must ensure the COPS classification complies with NFPA 110, Level 2 criteria, per NEC® Article 708. **(T-0)** COPS design, operational, and testing requirements are extensive and rigorous, requiring commitment of significant manpower and fiscal resources. AFCEC/CO is the approval authority for COPS classification. **Note:** AFRC installations submit to AFRC/A4C for final approval. ANG installations submit to NGB/A4IC CETB for processing.

*2.3.3. **Changed**. Others. Generators under 25 kW are not classified as RPIE or EAID and are

considered unit-owned equipment, and should not be maintained, tested, or operated by the BCE unless otherwise directed by command authorities; these generators fall under general purchase and acquisition policies.

*2.4.1. **Changed**. Redundant Generators. The mission owner must document the requirements for all new redundant generators, including any N+1, N+2, or N+X configurations, and AFCEC/COSM will verify these requirements before inclusion in programming documents (e.g., DD Form 1391, *FY__ Military Construction Project Data*). **(T-2)** An N+X configuration is a redundancy measure where N represents the number of assets (e.g., generators), required to power the mission and X is the number of backup assets. **Note:** AFRC installations submit to AFRC/A4C for final approval**.** ANG installations submit to NGB/A4IC CETB for processing.

* 2.4.2. **Changed**. Power Availability. All DAF power distribution systems need to deliver a minimum of 99.9% power availability during steady-state operation. This requirement does not apply to dedicated backup power systems. Document power availability for critical mission requirements of 99.9%, 99.99%, 99.999%, or 99.9999%. Alternatively, mission owners may provide allowable mission downtime expressed in time units (e.g., 30 seconds, 5 minutes, 4 hours). These availability levels require more than a generator; they include a mission-equipment power source to ensure continuous electrical service during power transfers. Power availability requirements are to be supported by mission-specific maximum outage durations (uptime and downtime) to economically design the standby power system.

*2.4.3. **Added**. Microgrids. All government-funded microgrids will be commissioned and revalidated every five years by a third-party contractor or AFCEC/COM, and documented via the Generator Authorization Tool on the CE DASH portal: https://usaf.dps.mil/sites/CE-DASH-Tools/GenAuthorization/Module/Home.aspx?page=Home. **(T-1)**

*2.4.3.1. **Added**. The BCE will report any changes to a microgrid's function or performance, differing from the conditions approved at time of installation, to AFCEC/COM via the Generator Authorization Tool in the CE DASH portal within 90 days. **(T-2)**

*2.4.3.2. **Added**. A microgrid will be considered part of the standby generator system when a central microgrid power production plant is installed and capable of islanding from the commercial power source; generator redundancy and power availability will be included in the analysis of the building's power system. **(T-1)**

***2.5. Added. Validation and Revalidation of Generators.**

*2.5.1. **Added**. The generator validation and authorization processes are to identify any changes to mission requirements or electrical system configurations. Ensure AF Form 487 data and monthly emergency runtime hours are properly entered in APIMS to support EPA air permitting. **Note**: AFRC installations submit to AFRC/A4C for final approval. ANG installations submit to NGB/A4IC CETB for processing.

*2.5.2. **Added**. Annual Revalidation.

*2.5.2.1. **Added**. Conduct and document revalidation of all generator designs, sizes, and configurations using the previous 12 months of AF Form 487 load data. Ensure each generator operates at or above 50% load capacity based on mission requirements (See **paragraph 2.6**). Record results in APIMS, input them into the Generator Authorization Tool within the CE DASH portal, and maintain supporting sizing records in APIMS.

*2.5.2.2. **Added**. Revalidations will be certified by the BCE using the standardized inventory template in the Generator Authorization Tool within the CE DASH portal. **(T-1) Note**: AFRC installations must be submitted to AFRC/A4C for final approval. ANG installations must be submitted to NGB/A4IC CETB for processing.

***2.6. Added. Generator Sizing, Configuration, and Design Approval.**

*2.6.1. **Added**. Process all generator sizing and configuration designs for RPIE and EAID generators through AFCEC/COSM in accordance with UFC 3-540-01. Maintain all authorizations in a generator record log. **(T-1)** Memorandum templates and fillable form examples are located on the CE DASH portal: https://usaf.dps.mil/sites/CE-DASH-Tools/GenAuthorization/Module/Home.aspx?page=Home. **Note**: AFRC installations must be submitted to AFRC/A4C for final approval. ANG installations must be submitted to NGB/A4IC CETB for processing.

*2.6.2. **Added**. Follow the formal design and sizing approval process for all generators to ensure they meet operational and configuration requirements. The BCE must review this process and submit it to AFCEC/COSM via the Generator Authorization Tool on the CE DASH portal. **(T-1)**

*2.6.3. **Added**. During the annual inventory or revalidation, verify generator sizing using test-load data from the previous 12 months and retain this data in the generator record. If mission-only load data does not demonstrate at least 50% load during normal operations, perform testing in accordance with the original equipment manufacturer's procedures. Report any oversized generators via the Generator Authorization Tool during the annual inventory and conduct an operational risk assessment. Oversized generators are prone to wet stacking; therefore, obtain approval from the mission owner and coordinate with the BCE.

*2.6.4. **Added**. When sizing a replacement or upstream generator, use AF Form 487 load data from the previous 12 months and upload this data to APIMS to support air permitting requirements. If a load bank is used to meet 50% threshold, annotate that portion separately on AF Form 487.

***3.1. Changed. Authorized Facilities.** Facilities listed in **paragraph 3.1.1** through **paragraph 3.1.34** are authorized generators as classified and indicated, based on a specified UFC or criteria in UFC 3-540-01. This authorization ensures efficient allocation of generators while accounting for limited funding, manpower, and local air permitting constraints. Report any backup power system failure to AFCEC/CO via the Generator Authorization Tool on CE DASH within 3 days, including the nature of the failure, the duration, and the resolution.

*3.1.33. **Added**. A single, centralized, government-only Electric Vehicle Charging Facility

(EVCF) per DAF installation [Other Permanently Installed].

*3.1.34. **Added.** A decentralized, government-only EVCF that is connected to a mission-critical facility authorized for backup power, as outlined in this chapter [Other Permanently Installed].

***Attachment 1**

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

*Added. UFC 3-540-01, *Engine-Driven Generator Systems for Prime and Standby Power Applications*, 01 August 2014

*Changed. DAFI 32-1001, *Civil Engineering Operations*, 27 February 2025

*Added. DAFI 32-9005, *Real Property Accountability*, 14 September 2022

*Added. DAFI 90-1701, *Installation Energy and Water Management*, 17 December 2020

*Changed. DAFMAN 91-203, *Air Force Occupational Safety, Fire, and Health Standards*, 25 March 2022

*Added. DoDM 4715.05, Volumes 1-5, *Overseas Environmental Baseline Guidance Document (OEBGD)*, 29 June 2020

Abbreviations and Acronyms

*Added. **AFCEC/CO—**Air Force Civil Engineer Center, Operations Directorate

*Added. **AFCEC/COM—**Air Force Civil Engineer Center, Operations Directorate, Maintenance Division

*Added. **AFCEC/COSM—**Air Force Civil Engineer Center, Operations Support Division, Electrical and Mechanical Branch

*Added. **AFSC—**Air Force Specialty Code

*Added. **APIMS—**Air Program Information Management System

*Added. **CA-CRL—**Custodian Authorization/Custody Receipt Listing

*Added. **CE DASH—**Civil Engineer Dashboard

*Changed. **CETB—**Civil Engineer Technical Branch (National Guard Bureau)

*Added. **FLDCOM—**Field Command

*Added. **IEP—**Installation Energy Plan

*Added. **MICT—**Management Internal Control Toolset (as in MICT SAC)

*Added. **NGB/A4IC CETB—**National Guard Bureau Civil Engineer Technical Branch

*Added. **SAC—**Self-Assessment Checklist (as in MICT SAC)

*Added. **SAF/IEE—**Deputy Assistant Secretary of the Air Force for Infrastructure, Energy, and Environment

*Added. **SMS—**Sustainment Management System

Added.** ***Terms

*Added. **Custodian Authorization/Custody Receipt Listing (CA-CRL)—**A formal document used to assign and track accountability for non-real property equipment issued to a custodian. It ensures assets are accurately recorded, maintained, and properly transferred or disposed of in accordance with DAF policy.

*Added. **CE DASH—**Air Force Civil Engineering Enterprise's technical support portal designed to provide resources and tools in a centralized online location. Access it here: https://usaf.dps.mil/teams/CEDASH/Scripts/Homepage/Home.aspx.

*Added. **Generator Authorization Tool—**A tool within the CE DASH portal used to manage the approval, inventory, validation, sizing, maintenance, and compliance documentation for all Department of the Air Force generators, ensuring alignment with UFC, EPA, and mission requirements. Access it here: https://usaf.dps.mil/sites/CE-DASH-Tools/GenAuthorization/Module/Home.aspx.

Administrative changes to AFMAN 32-1062, Electrical Systems, Power Plants and Generators

OPR: AF/A4CF

REPLACE "AFCEC/COSM" with "AFCEC/COM" (located at page 4, paragraph 1.7.2). 8 July 2021.

BY ORDER OF THE
SECRETARY OF THE AIR FORCE

AIR FORCE MANUAL 32-1062

20 OCTOBER 2020

CIVIL ENGINEERING

ELECTRICAL SYSTEMS, POWER PLANTS AND GENERATORS

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

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(Mr. John W. Henderson)

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This Air Force Manual (AFMAN) implements Air Force Policy Directive (AFPD) 32-10, *Installations and Facilities*. It provides acquisition and operation and maintenance (O&M) requirements for power systems, individual real property installed equipment (RPIE), and Equipment Authorized Inventory Data (EAID) generators. It applies to all civilian employees and uniformed members of the Regular Air Force, Air Force Reserve, and Air National Guard (ANG). Ensure all records created as a result of processes prescribed in this publication are maintained in accordance with AFMAN 33-322, *Records Management and Information Governance Program*, and disposed of in accordance with the Air Force Records Disposition Schedule located in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the Office of Primary Responsibility using the AF Form 847, *Recommendation for Change of Publication*; route AF Forms 847 from the field through the appropriate functional chain of command. This publication may be supplemented at any level, but all supplements must be routed to the Office of Primary Responsibility listed above for coordination prior to certification and approval. The authorities to waive wing/unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See Department of the Air Force Instruction 33-360, *Publications and Forms Management*, for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately, to the requestor's commander for non-tiered compliance items. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

SUMMARY OF CHANGES

This document has been substantially revised and must be completely reviewed. Major changes include revised guidance for generator authorizations and sizing approvals to reflect changes in Major Command (MAJCOM) roles and responsibilities. Guidance has also been updated for Environment Protection Agency requirements that impact air permitting. Responsibility has been assigned for third party funded contracts for projects that include prime power or micro-grid applications. The waiver authority for many requirements in this AFMAN has been reduced to reflect the need for installation-level ownership, management of risk, and commitment to executing Civil Engineer competencies. Engineering Technical Letter 13-4, Change 1: *Standby Generator Design, Maintenance, and Testing Criteria*, has been recinded.

Chapter 1

ROLES AND RESPONSIBILITIES

1.1. The Assistant Secretary of the Air Force Installations, Environment and Energy (SAF/IE). SAF/IE:

1.1.1. Establishes and issues policies for electrical systems, power plants, and generators that translate the ideas, goals or principles contained in the Air Force mission, vision and strategic plans.

1.1.2. Has overall responsibility for electrical systems, power plants, and generators to ensure the sustainability and operational readiness of the Department of the Air Force including, but not limited to, advocating for sufficient resources to sustain the mission.

1.1.3. Provides guidance, direction, and oversight for all matters pertaining to the formulation, review, and execution of plans, policies, programs, and budgets to ensure the sustainment and protection from mission degradation related to electrical systems, power plants, and generators.

1.1.4. SAF/IE or designee may review all T-1 and T-2 waiver decisions to ensure consistency with published policy, guidance, and approved Installation Energy Plans.

1.2. Air Force Deputy Chief of Staff for Logistics, Engineering and Force Protection, Directorate of Civil Engineers (AF/A4C). AF/A4C shall:

1.2.1. Oversee the electrical systems, power plants, and generators program to ensure compliance with applicable U.S. Codes, Federal Regulations, higher directives, and authorities and develop Management Internal Control Toolset self-assessment communicators.

1.2.2. Develop, maintain, clarify, and publish guidance for the electrical systems, power plants, and generators programs.

1.2.3. Oversee engagements with congressional representatives, congressional staff, and interdepartmental personnel with respect to electrical systems, power plants, and generators as delegated by SAF/IE.

1.3. Major Command Deputy Commanders (MAJCOM/CDs). shall approve deviations for semi-annual full-system testing, per **paragraph 1.11.3**

1.4. Air Force Reserve Command (AFRC), Installation Operations. AFRC shall:

1.4.1. Review and coordinate on generator authorization and design requests or recommended alternatives for AFRC installations. AFRC submits requests to Air Force Civil Engineer Center, Operations Directorate (AFCEC/CO) for approval, or disapproves submitted request, with no further action.

1.4.2. Review annual RPIE generator inventory and authorization reports, list of generators that are oversized, and those available for relocation provided by AFCEC Civil Engineer Maintenance, Inspection, and Repair Team (CEMIRT) (AFCEC/COM).

1.4.3. Coordinate with AFRC installations to redistribute available generators.

1.4.4. AFRC Headquarters (HQ) may delegate the tasks detailed in **paragraphs 1.4.1** through **1.4.3**

1.5. Office of the Director, Air National Guard (ANG). ANG shall:

1.5.1. Review and coordinate on generator authorization and design requests or recommended alternatives and submit requests to AFCEC/CO for approval, or disapprove with no further action.

1.5.2. Review annual RPIE generator inventory and authorization reports, list of generators that are oversized and available for relocation provided by AFCEC/COM CEMIRT.

1.5.3. Coordinate with ANG installations to redistribute available generators.

1.5.4. Office of the Director, ANG may delegate the tasks detailed in **paragraph 1.5.1** through **1.5.3**

1.6. Air Force Installation and Mission Support Center (AFIMSC). AFIMSC shall:

1.6.1. Review annual RPIE generator inventory and authorization reports, and list of generators that are oversized and available for relocation provided by AFCEC/COM CEMIRT. **Note:** This may not be delegated below the AFIMSC Detachment Commander level.

1.6.2. Coordinate with bases and AFCEC Mechanical and Electrical Division (AFCEC/COSM) to redistribute available generators. **Note:** This may not be delegated below the AFIMSC Detachment Commander level.

1.7. Air Force Civil Engineer Center. AFCEC shall:

1.7.1. Manage the approval process, in coordination with SAF/IE and AF/A4C.

1.7.1.1. Approve Generator authorization, design, and classification, to include Critical Operations Power Systems (COPS)

1.7.1.2. Revalidate annual generator authorizations see (**paragraph 1.9.6**).

1.7.1.3. Approve use of generators for prime power, cogeneration, or micro-grid applications.

1.7.2. Maintain the master list of RPIE generators available for relocation as compiled from AFCEC/COSM, and shall forward a copy to Air Force Installation and Mission Support Center/Installation Engineering Division (AFIMSC/IZB).

1.7.3. Provide guidance regarding federal, state, and local environmental compliance, including, but not limited to, air quality, spill containment, and fuel tank inspection requirements.

1.7.4. Provide an enterprise-level compliance management system (i.e., Air Program Information Management System) and maintain air quality compliance guidance (e.g., operations, maintenance, repair, and replacement of engines).

1.7.5. Review and coordinate on use of generators for co-generation, prime power, renewable, or micro-grid applications. The AFCEC Energy Directorate (AFCEC/CN) is the approval authority for all projects funded by third party contracts for co-generation, prime power, renewable, or micro-grid applications using energy cost savings. AFCEC/CN has

responsibility for certifying operational third party funded co-generation, prime power, renewable, or micro-grid applications for the life of third party contracts.

1.7.6. Coordinate authorization and design approval requests to AFCEC/CO when projects are programmed without generator authorization.

1.8. Installation Commander (or Equivalent). shall, as the Responsible Official, be the approving authority for deviations from manufacturer service intervals and ensure compliance with all applicable air quality permits. **(T-0).** The list of applicable laws includes, but is not limited to: Title 40 Code of Federal Regulations (CFR), Part 63 Subpart ZZZZ, *National Emissions Standards for Hazardous Air Pollutants for Stationary Reciprocating Internal Combustion Engines*; 40 CFR Part 60 Subpart JJJJ, *New Source Performance Standards (NSPS) - Standards of Performance for Stationary Spark Ignition Internal Combustion Engines* (the "spark ignition NSPS rule"); 40 CFR Part 60 Subpart IIII, *Standards of Performance for Stationary Spark Ignition Internal Combustion Engines* (the "Compression Ignition NSPS rule"); 40 CFR Part 1039, *Control Of Emissions from New and In-Use Nonroad Compression-Ignition Engines*; 40 CFR Part 89, *Control of Emissions from New and In-Use Nonroad Compression-Ignition Engines*; 40 CFR Part 90, *Control of Emissions From Nonroad Spark-Ignition Engines at or Below 19 Kilowatts*; 40 CFR Part 1048, *Control of Emissions from New Large Nonroad Spark-Ignition Engines*.

1.9. The Base Civil Engineer (BCE). BCEs shall:

1.9.1. Provide, operate, and maintain all real property and RPIE electrical power systems and related items, including EAID assigned to the BCE and RPIE items, with the exception of units supporting missile systems or special-use tenant generators lacking a signed memorandum of agreement (MOA). **(T-1).** The BCE will ensure preventive maintenance procedures comply with AFI 32-1001, *Civil Engineer Operations*. **(T-1).** The BCE will also ensure testing frequencies comply with Unified Facilities Criteria (UFC) 3-540-07, *Operation and Maintenance (O&M): Generators*. **(T-0).** The BCE is responsible for operating and maintaining other real property and RPIE electrical systems (as applicable to each installation), which include:

1.9.1.1. Controls, sensors, and alarm circuits needed for operation of real property facilities, such as tank liquid level sensors and alarms; fire alarm systems; mass notification systems; facility-integrated photovoltaic panels, arrays, and components; joint-service interior intrusion detection systems; electrical heating and air-conditioning systems using equipment similar to RPIE; Electrical systems temporarily deployed during exercises or contingency or wartime operations, or systems permanently assigned to an installation. **(T-2).**

1.9.1.2. Systems for utility plant management and distribution, such as energy management and control systems, which include supervisory control and data acquisition systems and utility monitoring and control systems. The BCE will obtain advance written approval from AFCEC/COSM to install any generator security device, such as those used to mitigate electrical system vulnerability. **(T-2).**

1.9.2. Ensure required MOAs are established; that they address reimbursable/non-reimbursable expenses and requests for additional levels of maintenance; and that they receive Judge Advocate review. **(T-2).** The BCE has no maintenance responsibility for

electrical equipment that is not on the real property record or civil engineer (CE) allowance standard and has no MOA in place. The BCE will maintain and test electrical power systems at special-use or tenant facilities in accordance with the signed MOA and Air Force standards for RPIE and equipment typically assigned to the CE squadron. **(T-2).** Using organizations are responsible for operator-level maintenance of equipment supplying power exclusively to these facilities when no MOA is in place. Real property accountability for equipment supplying power to a special-use facility or tenant organization resides with the BCE. **(T-3).**

1.9.2.1. The BCE will review and approve non-RPIE generator connections requested by tenant organizations through the work order process per **paragraph 1.10** **(T-3).**

1.9.2.2. Defense Logistics Agency capitalized real property and generator operation and maintenance that is funded through the Defense Logistics Agency Sustainment, Restoration and Modernization process is not a BCE responsibility without an MOA in place. If a MOA is in place, the terms of the MOA will determine BCE responsibility. **(T-3)**.

1.9.2.3. For Medical/Dental facilities funded through the Defense Health Program Sustainment, Restoration and Modernization program, generator operations and maintenance are not a BCE responsibility without an MOA in place. If a MOA is in place, the terms of the MOA will determine BCE responsibility, and BCE will be obligated to comply with those terms. **(T-2)**. Generator authorization, design, classification and revalidation for medical/dental facilities will be accomplished by the Defense Health Agency.

1.9.3. Develop and maintain as-built electrical system drawings, operating procedures, schematics, manufacturer O&M manuals, and emission certifications. **(T-3).** The BCE will ensure a copy of local operating procedures, a list of authorized operators, and a one-line diagram are posted at each site. **(T-3).**

1.9.4. Develop and coordinate design requests and authorizations for RPIE and BCE-owned EAID generators through AFCEC, including generators procured with military construction, O&M, sustainment, restoration, and modernization, Defense Logistics Agency, or other funding. **(T-2).** All authorizations will be maintained in a generator record log. **(T-2).** Memorandum templates and fillable form examples are located on the CE DASH site: **https://cs2.eis.af.mil/sites/10159/Shared%20Documents/Forms/AllItems.aspx**. **Note:** ANG installations submit to National Guard Bureau (NGB) Civil Engineer Technical Service Center (CESTC) (NGB/A4OC) for processing and forwarding to AFCEC.

1.9.5. Conduct an annual inventory of all CE-operated and maintained RPIE and EAID generators by 1 August each year, including generators that are replacement-eligible, and forward a copy to AFCEC/COSM. **(T-2).** RPIE generators shall be reconciled with real property records and EAID generators with Custodian Authorizations and Custody Receipt Listing records. **(T-2). Note:** ANG installations submit to NGB/A4OC CESTC for processing and forwarding to AFCEC.

1.9.6. Conduct a validation of generator authorizations annually to identify changes to mission or electrical system configurations, and forward revalidation reports to AFCEC/COSM for revalidation. **(T-2).** Revalidations must be certified by the BCE in the form of an inventory list that includes building locations, mission supported, and any changes

in mission or loads based on data recorded on AF Form 487, *Generator Operating Log (Inspection Checklist)*, from the previous year. **(T-3)**. (If the unit is using Air Program Information Management System as the official shop record, Air Program Information Management System will serve as the validation platform for authorization). **Note:** ANG installations submit to NGB/A4OC CESTC for processing and forwarding to AFCEC.

1.9.7. Ensure compliance with all applicable U.S. federal, state, and local requirements (including permits); or, for overseas, the Final Governing Standard (or Overseas Environmental Baseline Guidance Document where no Final Governing Standard exists). **(T-0).**

1.9.8. Develop written prioritized refueling plans and requirements to support generator operation (EAID and RPIE and other generator operation identified within an approved MOA) during extended power outages of at least seven days. **(T-3).** All combustible fuels must comply with applicable state and federal fuel standards. **(T-1).** Liquid propane gas and natural gas are permitted only in accordance with UFC 3-540-07, Appendix C. **(T-1).**

1.9.9. Establish an onsite fire extinguisher near the generator in accordance with AFMAN 91-203, *Air Force Occupational Safety, Fire, and Health Standards*. **(T-3).** This shall be compliant with all applicable safety codes and standards such as NFPA 70, *National Electrical Code®*, National Fire Protection Association (NFPA) 70E, *Standard for Electrical Safety in the Workplace®*, AFMAN 32-1065, *Grounding and Electrical Systems*, UFC 3-560-01, *Operation and Maintenance: Electrical Safety*, and all current electrical safe work practices in accordance with 29 CFR Part 1910, *Occupational Safety and Health Standards*. **(T-0).**

1.9.10. Account for generators in accordance with UFC 3-540-07. **(T-2).**

1.9.11. Provide training in generator operation to Facility Managers when requested, scheduled during monthly generator testing. **(T-3).**

1.10. Installation Management Flights (Facility Manager for Air National Guard). Installation Management Flights shall coordinate with the BCE prior to generator installations and relocations to ensure compliance with guidelines (at least 90 days prior to scheduled installation to provide adequate time for permitting and notification requirements). **(T-3).**

1.11. Installation Facility Managers. Facility managers shall:

1.11.1. Verify proper function of all building equipment and systems during generator functional testing using facility load and sign the AF Form 487 (or electronic version) following the semi-annual generator test. **(T-3).**

1.11.2. Maintain and post next to the generator a list of facility personnel trained to operate facility generators when generator operation by facility personnel is requested. **(T-3)**. User training is valid for one year only and must be requested through and approved by the BCE. **(T-3).**

1.11.3. Submit written request to the BCE for MAJCOM/CD approval to extend the semi-annual generator test. **(T-2).** Request with justification must be submitted at least 90 days prior to the semi-annual generator test. **(T-3)**.

1.11.4. Provide an organizational memorandum for record for facility electrical system testing and maintenance that is deferred for more than six (6) months. **(T-2).**

1.12. Tenant Organizations. All tenant organizations shall submit a work request for any generator system to be connected to real property, and establish a MOA with the BCE for the operation and maintenance of the generator on a reimbursable basis. **(T-2).** Tenant organizations lacking a signed MOA transferring electrical systems responsibility to the BCE must satisfy all requirements of this instruction organically or through unit funded, contract maintenance services. **(T-1).**

Chapter 2

AUTHORIZATION, CLASSIFICATION, AND CONFIGURATION OF GENERATORS.

2.1. Prime Power. AFCEC/CO manages, in coordination with the office of the Deputy Assistant Secretary of the Air Force for Environment, Safety, and Infrastructure (SAF/IEE) and AF/A4C, the use of generators for prime power, micro-grids, or cogeneration. When submitting a request for cogeneration or micro-grids within the area electrical power system, the BCE must demonstrate that the system complies with Institute of Electrical and Electronics Engineers (IEEE) 1547, *Standard for Interconnecting Distributed Resources with Electric Power Systems*, and negotiate an interconnection agreement with the area electrical power system (utility) supplier. **(T-2).** Coordination with the BCE's Environmental Section is required prior to authorization for purposes of regulatory permitting and notifications. **(T-3).**

2.2. Dedicated Standby Generators. The BCE will ensure that authorized mission-essential functions are supported by dedicated standby generators. **(T-3)**. Generators authorized to support mission-essential functions are installed and connected to provide power only to mission-essential functions and any critical facility infrastructural support systems within a single facility in the event there is a loss of commercial power. If unique circumstances exist where one standby system is required to support multiple facilities, the BCE (or appropriate commander) will submit an authorization request to AFCEC/CO for approval. **(T-2)**. Installation Energy Plans or other energy resilience planning may identify opportunities to interconnect

multiple standby generators and multiple facilities; in such cases, SAF/IEE may approve projects based on the concept of interconnecting multiple standby generators and multiple facilities.
Note: ANG installations submit to NGB/A4OC CESTC for processing and forwarding to AFCEC.

2.3. Types of Standby Generators. Three types of generators may be authorized: RPIE, EAID, and other.

2.3.1. RPIE Generators. Support mission-critical functions where controlled shut-down or delayed power restoration is unacceptable. They may be authorized for missions requiring immediate power restoration, uninterrupted power, or support for emergency systems as defined in National Electrical Code® (NEC®) Article 700, *Emergency Systems*. RPIE generators are classified as "Emergency," "COPS," "Other Permanently Installed," or "Petroleum, Oils, and Lubricants (POL)/Fuels".

2.3.1.1. "Emergency." This classification applies to life safety systems required to comply with NEC® Article 700. The BCE must ensure emergency systems also comply with NFPA 110, *Standard for Emergency and Standby Power Systems*, Level 1 criteria. **(T-1).**

2.3.1.2. "COPS." The BCE must ensure the "COPS" classification (applies to NEC® Article 708, *Critical Operations Power Systems [COPS]*) complies with NFPA 110, Level 2 criteria. COPS design, operational, and testing requirements are extensive and rigorous, requiring commitment of significant manpower and fiscal resources. **(T-1).** AFCEC/CO is the approval authority for COPS classification. **Note:** ANG installations submit to NGB/A4OC CESTC for processing and forwarding to AFCEC.

2.3.1.3. "Other Permanently Installed." This classification applies to other mission-critical functions not designated as "Emergency" or "COPS." The BCE must ensure these systems comply with NEC® Article 701, *Legally Required Standby Systems*, and NFPA 110 Level 2 criteria. **(T-1).**

2.3.1.4. "POL/Fuels". This classification applies to Fuels Information Service Centers (Fuels Operation) and Types III, IV, and V hydrant fueling systems designed in accordance with Department of Defense Standard Designs AW 078-24-28, *DOD Pressurized Hydrant Fueling System Type III*, and AW 078-24-29: *DOD Standard Pressurized Hydrant Direct Fueling System Type IV/V*, with a manual interlocked transfer switch. **(T-0).** The BCE must ensure these systems are sized to 50 percent of pumping capacity within the continental United States and 100 percent of pumping capacity outside the continental United States. **(T-2).**

2.3.2. EAID Portable Units. Either trailer-mount or skid-mount generators are classified as either "POL/Fuels" or "Portable".

2.3.2.1. "POL/Fuels." The BCE must ensure the EAID "POL/Fuels" classification is applied to Types I and II hydrant fueling systems, bulk fuel storage areas, non-hydrant operating storage areas, and vehicle fueling stations that are prepositioned. **(T-2).** The BCE must ensure that detached fuels testing labs not located in a fuels operations facility have a generator connection plug and a manual transfer switch. **(T-2).**

2.3.2.2. "Portable." These generators may be authorized for mission-essential functions where delayed power restoration is acceptable and portable generation decreases simultaneous risk to the facility and generator.

2.3.3. Others. Generators (5 kilowatts or less) not classified as RPIE or EAID are not addressed in this manual. These generators are unit-owned equipment items, are not the responsibility of the BCE to maintain, test, or operate, and are subject to purchase and acquisition policies generally applicable to equipment.

2.4. Generator configuration and Redundancy.

2.4.1. Redundant Generators. Requirements for new redundant generators must be documented by the mission owner and verified by AFCEC/CO before being included in programming documents (e.g., DD Form 1391, *FY _____ Military Construction Project Data*). **(T-2)**. This includes any of the N+1, N+2, and N+X configurations. **(T-2).** An N+X configuration is a redundancy measure where N represents the number of assets, for instance generators, required to power the mission and X is the number of backup assets. **Note:** ANG installations submit to NGB/A4OC CESTC for processing and forwarding to AFCEC.

2.4.2. Power Availability. Critical missions must also have documented power availability requirements of 99.9%, 99.99%, and 99.999%. **(T-2).** Alternatively, mission owners can also provide their annual "allowable mission downtime" expressed in units of time (i.e. 30 sec, 5 min, 4 hrs, etc.). These levels require more than just a generator and must have a mission-equipment power source to continue electrical service during power source transfer. These power availabilities must also have mission-specific maximum power outage times (i.e., uptime and downtime) to economically design the standby power system.

Chapter 3

AUTHORIZATION AND CLASSIFICATION OF FACILITIES

3.1. Facilities . Listed in **paragraphs 3.1.1** through **3.1.31** will be authorized generators and classified as listed below, per the specified UFC or the criteria in UFC 3-520-01, *Interior Electrical Systems*. **(T-0).** Other facilities not addressed by these UFCs must be submitted to AFCEC/CO for appropriate level approval. **(T-2).**

3.1.1. Medical healthcare and ambulatory care facilities (excludes medical business occupancies as defined by NFPA) in accordance with UFC 4-510-01, *Design: Medical Military Facilities.* **(T-2).**

3.1.2. Air navigation aids and facilities, and airfield lighting COPS or Other Permanently Installed. **(T-2).**

3.1.3. Mission/emergency-essential refrigerated storage rooms [Portable]. **(T-2).**

3.1.4. POL storage and dispensing facilities [POL/Fuels RPIE or EAID]. **(T-2).**

3.1.5. Mission-essential/critical utility plants [Other Permanently Installed]. **(T-2).**

3.1.6. Critical sewage lift stations with low-level to high-level alarm duration less than EAID deployment time [Other Permanently Installed]. **(T-2).**

3.1.7. Typical sewage lift stations with adequate alarm notification to start pre-positioned generator [Portable or Other Permanently Installed when required by local laws]. **(T-2).**

3.1.8. Civil Engineer control centers [Portable or Other Permanently Installed]. **(T-2).**

3.1.9. Mission-essential/critical communications facilities, information transfer nodes, and telephone exchanges [Other Permanently Installed]. **(T-2).**

3.1.10. Fire stations, including fire alarm, fire control, and radio equipment [Emergency]. **(T-2).**

3.1.11. Mission-essential computer automated data processing facilities [Other Permanently Installed]. **(T-2).**

3.1.12. Air traffic control towers [COPS or Other Permanently Installed]. **(T-2).**

3.1.13. Base weather stations [Other Permanently Installed or Portable]. **(T-2).**

3.1.14. Surveillance and warning facilities, to include ground-based space telescopes [Other Permanently Installed]. **(T-2).**

3.1.15. Primary command and control facilities (does not include headquarters facilities without direct and essential command and control functions) [Other Permanently Installed]. **(T-2).**

3.1.16. Remotely piloted aircraft control pods that are hardwired into a RPIE facility [Other Permanently Installed or Portable]. **(T-2).**

3.1.17. Munitions storage facilities [Other Permanently Installed or Portable]. **(T-2).**

3.1.18. Entry control points, security gates, and related security lighting systems [Portable]. **(T-2).**

3.1.19. Aircraft and aircrew alert facilities [Other Permanently Installed]. **(T-2).**

3.1.20. Law enforcement and security facilities [COPS, Other Permanently Installed, or Portable]. **(T-2).**

3.1.21. Emergency operations centers in accordance with UFC 4-141-04, *Emergency Operations Center Planning and Design* [Other Permanently Installed]. **(T-2).**

3.1.22. Mission, property, and life support facilities at remote and not readily accessible sites, such as split-site aircraft warning and surveillance installations [Other Permanently Installed]. **(T-2).**

3.1.23. One dining facility per installation or geographic location [Other Permanently Installed or Portable]. **(T-2).**

3.1.24. Industrial facilities that have noxious fumes requiring removal (provide power for the exhaust system only; in the case of an aircraft corrosion control facility, also provide power to the hangar doors in accordance with UFC 4-211-02, *Aircraft Corrosion Control and Paint Facilities*) [Emergency]. **Note:** Aircraft fuel cell repair facilities are not authorized standby generators. **(T-2).**

3.1.25. Readiness facilities relying on electrical power to support tactical or mission-essential operations or facilities uniquely required on the base Contingency Response Plan [Other Permanently Installed]. **(T-2).**

3.1.26. Intelligence processing facilities providing mission-essential support to combat and contingency tactical missions [Other Permanently Installed]. **(T-2).**

3.1.27. Simulation or materials laboratories where continuous power is needed for human safety or to maintain low-tolerance temperature (<5 degrees Fahrenheit) and humidity (<5 percent) control to avoid catastrophic consequences [Other Permanently Installed]. **(T-2).**

3.1.28. Emergency lighting, elevators, fire alarms, security systems, or other facility life safety equipment within high-occupancy buildings or places of assembly of 1,000 or more people, for safely moving people out [Emergency]. **(T-2).**

3.1.29. Space launch/lift, vertical integration facilities [Other Permanently Installed or EAID].

3.1.30. Space command and control, satellite, and cyber operations facilities [Other Permanently Installed or COPS]. **(T-2).**

3.1.31. As required by other Department of Defense component, Federal branch, or U.S. state. **(T-2).**

3.1.32. Task Critical Assets and Critical Assets (formerly known as supporting infrastructure critical assets) in accordance with AFI 10-2402, *Critical Asset Risk Management Program* [Other Permanently Installed or COPS]. **(T-2).** (examples: generators for submarines, National Aeronautics and Space Administtration (NASA), National Reconaissance Office (NRO), Missile Defense Agency (MDA), state tenant on Air Force installation).

WARREN D. BERRY, Lieutenant General, USAF

DCS/Logistics, Engineering & Force Protection

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

29 CFR Part 1910, *Occupational Safety and Health Standards*

40 CFR Part 60 Subpart IIII, *Standards of Performance for Stationary Compression Ignition Internal Combustion Engines*, 1 July 2016

40 CFR Part 60 Subpart JJJJ, *Standards of Performance for Stationary Spark Ignition Internal Combustion Engines*, 1 July 2016

40 CFR Part 63 Subpart ZZZZ, *National Emissions Standards for Hazardous Air Pollutants for Stationary Reciprocating Internal Combustion Engines*

40 CFR Part 89*, Control of Emissions from New and In-Use Nonroad Compression-Ignition Engines,* 17 June 17 1994

40 CFR Part 90*, Control of Emissions From Nonroad Spark-Ignition Engines at or Below 19 Kilowatts,* July 3, 1995

40 CFR Part 1039, *Control Of Emissions from New and In-Use Nonroad Compression-Ignition Engines*, June 29, 2004

40 CFR Part 1048, *Control of Emissions from New Large Nonroad Spark-Ignition Engines*, 8 November 2002

UFC 3-520-01, *Interior Electrical Systems*, 6 October 2015

UFC 3-540-07, *Operation and Maintenance (O&M): Generators*, 21 February 2018

UFC 3-560-01, *Operation and Maintenance: Electrical Safety*, 21 February 2018

UFC 4-141-04, *Emergency Operations Center Planning and Design,* 15 July 2008

UFC 4-211-02, *Aircraft Corrosion Control and Paint Facilities,* 12 December 2012

UFC 4-510-01, *Design: Medical Military Facilities*, 1 November 2017

AFPD 32-10, *Installations and Facilities*, 20 July 2020

AFI 10-2402, *Critical Asset Risk Management Program*, 29 August 2017

AFI 32-1001, *Civil Engineer Operations,* 4 October 2019

AFI 33-322, *Records Management and Information Governance Program*, 23 March 2020

AFMAN 32-1065, *Grounding and Electrical Systems,* 17 July 2020

AFMAN 91-203, *Air Force Occupational Safety, Fire, and Health Standards*, 11 December 2018

DAFI 33-360, *Publications and Forms Management,* 1 December 2015

NEC® Article 700, *Emergency Systems*

NEC® Article 701, *Legally Required Standby Systems*

NEC® Article 708, *Critical Operations Power Systems [COPS]*

NFPA 70, *National Electrical Code*® (NEC®)

NFPA 70E, *Standard for Electrical Safety in the Workplace*

NFPA 110, *Standard for Emergency and Standby Power Systems*

IEEE 1547, *Standard for Interconnecting Distributed Resources with Electric Power Systems*

DoD Standard Design AW 078-24-28, *DOD Pressurized Hydrant Fueling System Type III,* 1 March 2020

DoD Standard Design AW 078-24-29*, DOD Standard Pressurized Hydrant Direct Fueling System Type IV/V,* 1 July 2010

Adopted Forms

DD Form 1391, *FY _____ Military Construction Project Data*

AF Form 847, *Recommendation for Change of Publication*

AF Form 487, Generator Operating Log (Inspection Checklist)

Abbreviations and Acronyms

AFCEC—Air Force Civil Engineer Center

AFIMSC—Air Force Installation and Mission Support Center

AFMAN—Air Force Manual

AFRC—Air Force Reserve Command

ANG—Air National Guard

BCE—Base Civil Engineer

CE—Civil Engineer

CEMIRT—Civil Engineer Maintenance, Inspection, and Repair Team

CESTC—Civil Engineer Technical Service Center

CFR—Code of Federal Regulations

COPS—Critical Operations Power System

DAFI—Department of the Air Force Instruction

EAID—Equipment Authorized Inventory Data

EPA—Environmental Protection Agency

HQ—Headquarters

IEEE—Institute of Electrical and Electronics Engineers

MAJCOM—Major Command

MAJCOM/CD—MAJCOM Deputy Commander

MOA—Memorandum of Agreement

NEC®—National Electrical Code

NFPA—National Fire Protection Association

NGB—National Guard Bureau

NSPS—New Source Performance Standards

O&M—Operations and Maintenance

POL—Petroleum, Oils, and Lubricants

RPIE—Real Property Installed Equipment

UFC—Unified Facilities Criteria